Financial Cycles with Heterogenous Intermediaries

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Macro reading group

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Motivation

- GFC 2007-8: the financial sector drives fluctuations in economic activity and aggregate risk.
- Crises preceded by rapid expansion of debts: *Booms gone bust*

  [Fisher, 1933, Minsky, 1984, Kindleberger, 1978, Schularick and Taylor, 2012, Jordà et al., 2013]
- Shock amplification and propagation with financial frictions

  [Kiyotaki and Moore, 1997, Bernanke et al., 1999, Brunnermeier and Sannikov, 2014, Shin, 2009]

This paper

- The boom phase of the cycle: endogenous risk build-up
- Rich financial sector embedded within a DSGE model
- Heterogeneity in risk-taking attitudes (VaR)
- Limited liability

Overview of results

- Endogenous non-linear trade-off between monetary policy and financial stability.
  - High interest rates $\longrightarrow$ rate cuts lead to the entry of less risk-taking intermediaries
  - Low interest rates $\longrightarrow$ higher concentration of risk among riskier intermediaries
- Different financial structures can underly the same aggregate outcome
- "Good booms" vs. "bad booms"
  - Low funding costs $\longrightarrow$ high risk-shifting, lower risk premia
  - Positive productivity shock $\longrightarrow$ maintain financial stability and risk premia

Which state variable?

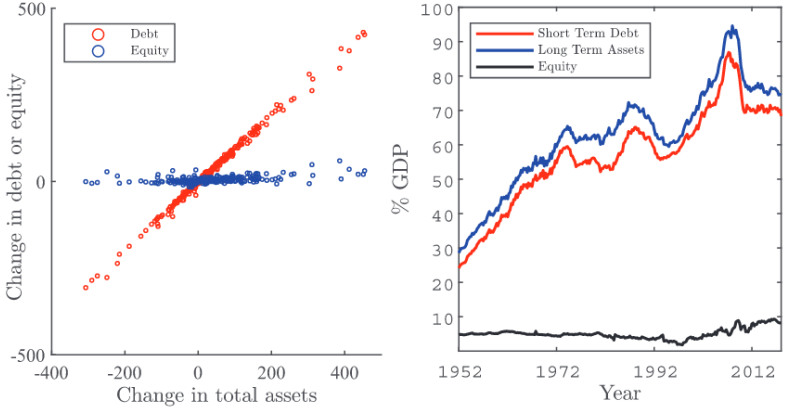


FIGURE A3

Left panel: Yearly changes in total asset against yearly changes in equity or debt from 1993 to 2015. Billions of USD. Source: Bankscope. Right panel: Bank short-term debt, long-term assets and equity as a percentage of US GDP. Data constructed as in Krishnamurthy and Vissing-Jorgensen (2015)

Heterogeneity across which dimension?

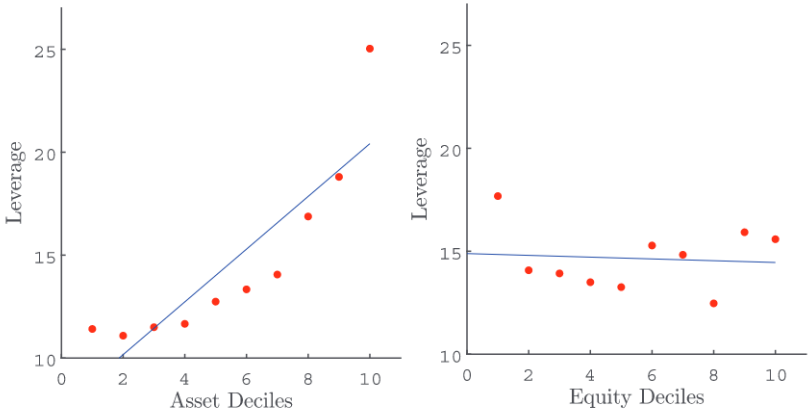


FIGURE 2

Mean leverage by asset deciles (left) and equity deciles (right)

Note: Binned scatter plots with intermediaries grouped in the x-axis into 10 bins by balance sheet size (left panel) and equity (right panel). Dots represent the mean leverage (unweighted) of each group. Each bin contains an average of 41 distinct financial intermediaries and 606 observations.

Sensitivity to the cycle

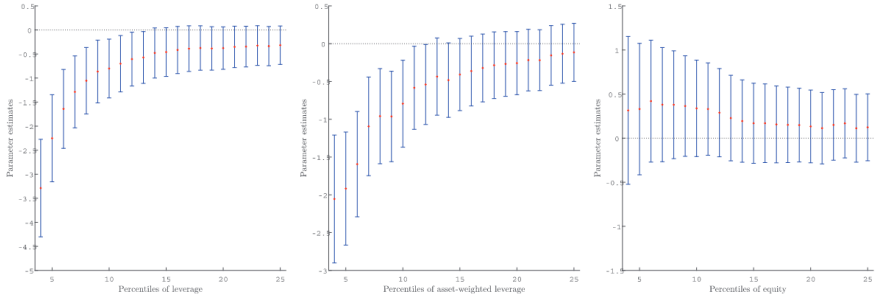


FIGURE 4
Interaction coefficients under different percentile choices

Note: Point-estimates and 95% significance bounds for θ_3 in the following regression: $Lev_{i,t} = \theta_0 + \theta_1 Lev_{i,t-1} + \theta_2 TopX^j_{i,t} + \theta_3 FF_t \times TopX^j_{i,t} + \nu_i + \mu_t + \varepsilon_{i,t}$. $TopX^j_{i,t}$ is a dummy variable, that takes the value 1 if the intermediary has leverage (left graph, $j = \{\}$), asset-weighted leverage (middle graph, $j = \{AW\}$) or equity (right graph, $j = \{Eq\}$) *above* the X-axis percentile value for year t.

Stylized facts and objective

1. The larger intermediaries are the most levered
2. Concentration of assets goes up as costs of funds go down
3. Higly levered intermediaries are more sensitive to changes in the cost of funds

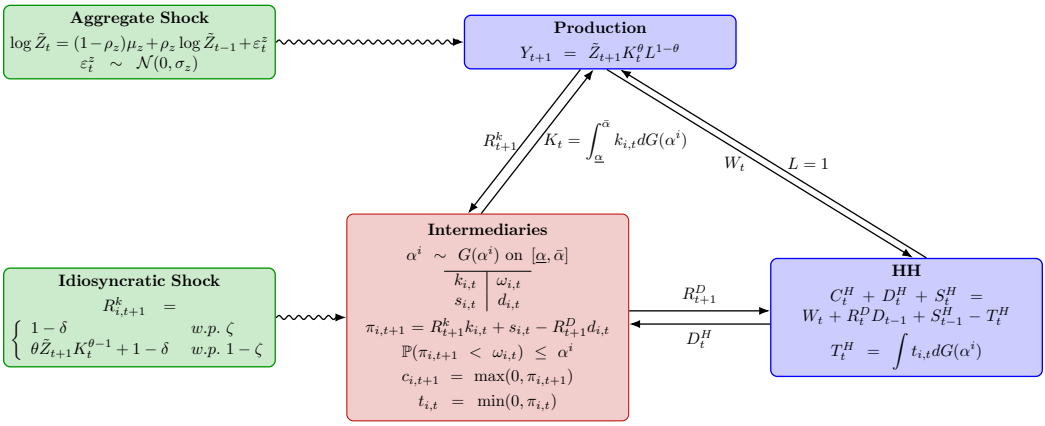
Goal $\longrightarrow$ rationalize these features within a dynamic stochastic model

Key ingredient: intermediary specific Value-at-Risk (VaR) constraint:

$$VaR = \inf_{V \geq 0} \{\mathbb{P}(A \leq A_0 - V) \leq \alpha^i\}$$

- Regulatory (Basel II) and internal tool
- Rationalizes procyclical leverage (Adrian and Shin, 2014)

Outline of the model



Add wholesale funding and monetary policy shocks afterwards

Intermediary investment problem

OLG structure $\longrightarrow$ start with $\omega_{i,t} = \omega$ and consume the resulting net worth.

Deposit insurance $\longrightarrow$ $\pi_{i,t+1}$ truncated at 0 $\longrightarrow$ Option Value of default

Storage technology available $\longrightarrow$ Outside option

Three decisions:

1. Participation $\longrightarrow$ k or s?
2. If $k \longrightarrow \lambda = \frac{k}{\omega} \geq 1$?
3. If $\lambda > 1 \longrightarrow \lambda =$?

Intermediary investment problem

Answering 3 is easy : provided $\mathbb{E}_t(R^k_{t+1}) > 1$, then if $\lambda_{i,t} > 1$ the VaR condition binds:

$$\mathbb{P}(\pi_{i,t+1} \leq \omega_{i,t}) = \alpha^i \iff \lambda_{i,t} = \frac{r^D_t}{r^D_t - \theta Z^e_{t+1} K_t^{\theta-1} F^{-1}\left(\frac{\alpha^i - \zeta}{1-\zeta}\right) + \delta} \tag{1}$$

- Heterogenous intensive margin: $\frac{\partial^2 \lambda_{i,t}}{\partial (r^D_t)^2} > 0$, $\frac{\partial^2 \lambda_{i,t}}{\partial r^D_t \partial \alpha^i} < 0$: $\downarrow r_d \longrightarrow \uparrow$ market share of the most risk-taking intermediaries. The lower r^D, the stronger the composition effect and the stronger the intensive margin.
- Volatility paradox: $\frac{\partial \lambda_{i,t}}{\partial \sigma_z} < 0$

Intermediary investment problem

Answering 1 : participation ?

- A participating but safe intermediary has a risk budget given by

$$F\left(\frac{\delta K_t^{1-\theta}}{\theta Z_{t+1}^e}\right) \leq \frac{\alpha^i - \zeta}{1-\zeta}$$

- This does not depend on $k_{i,t}$
- If this inequality is true, $k_{i,t} = \omega$, if not $s_{i,t} = \omega$

The marginal intermediary for which this holds with equality characterizes α^N:

$$\alpha_t^N = \zeta + (1-\zeta)F\left(\frac{\delta K_t^{1-\theta}}{\theta Z_{t+1}^e}\right) \tag{2}$$

Intermediary investment problem

Answering 2: to lever-up or not to lever-up?

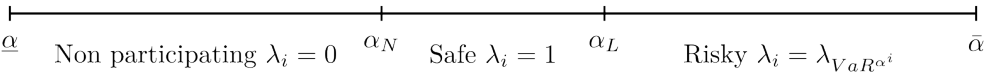
Since a participating intermediary invests at least ω, the marginal intermediary j satisfies:

$$\underbrace{\mathbb{E}_t\left[\max(0, k_{j,t}(R^k_{j,t+1} - R^D_t) + R^D_t\omega)\right]}_{V^L_{j,t}} = \underbrace{\omega\mathbb{E}_t R^k_{t+1}}_{V^N_{j,t}} \tag{3}$$

(1)+(3) define a cutoff value $\alpha^L_t = \alpha^L_t(r^D_t, Z^e_{t+1}, K_t)$

Financial Market Equilibrium

Three cases :



- $\alpha_t^L \longrightarrow$ Extensive margin + measure of financial stability
- $\int_{\alpha^L}^{\bar{\alpha}} \lambda_i dG(\alpha^i) \longrightarrow$ Intensive margin

Equilibrium:

$$K_t = \int_{\alpha_t^L}^{\bar{\alpha}} k_{i,t} dG(\alpha_i) + \left[G(\alpha_t^L) - G(\alpha_t^N)\right] \omega \tag{4}$$

(1)+(3)+(2)+(4) determine $K_t(r_t^D, Z_{t+1}^e)$

Comparative statics

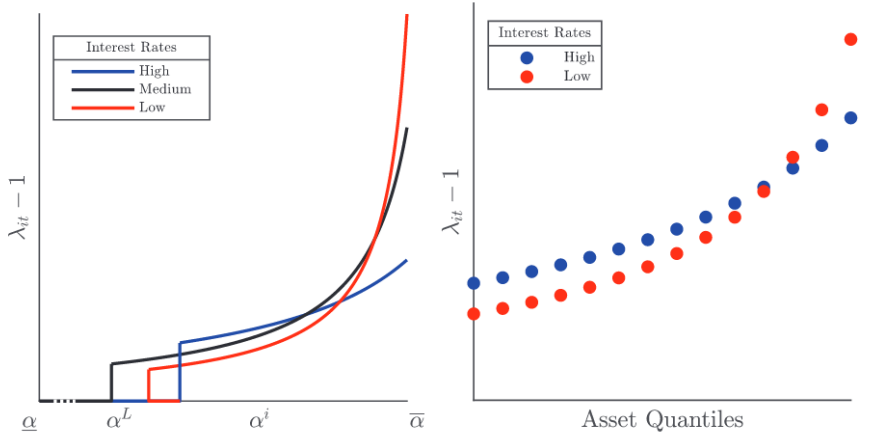


FIGURE 5

Cross-sectional distribution of leverage λ_{it} as a function of the VaR parameter α^i (left panel) and quantiles of asset size among leveraged institutions (right panel). $\lambda_{it} - 1$ is used so that unlevered participants have a value of 0.

The seeds of the trade-off

Non-linear (state-dependant) effect: a lower r_t^D has two effects:

1. $\downarrow$ RoL $\longrightarrow$ relaxes the VaR constraint $\longrightarrow$ expansion + entry
2. Pecuniary externality: $\uparrow K$ leads to $\downarrow R^k \longrightarrow$ exit

If r^D is low, risky intermediaries expand a lot when $\downarrow r_d$ which leads to a strong pecuniary externality : less risky intermediary are "priced-out" of the market

$$\frac{\partial \alpha^L}{\partial r}(\lambda^{\alpha^L} - 1) = \int_{\alpha^L}^{\bar{\alpha}} \frac{\partial \lambda^\alpha}{\partial r} dG(\alpha) - \frac{\partial K}{\partial r}$$

Non-monotonic financial structure

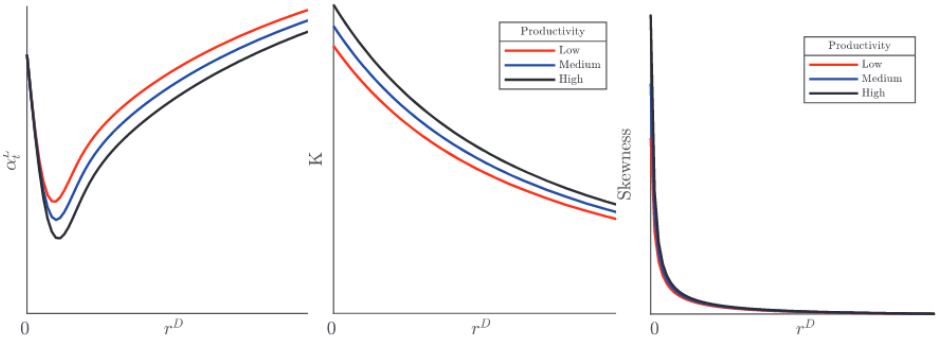


FIGURE 6

Cut-off level α_t^L and aggregate capital stock as a function of deposit rates r_t^D

Financial stability vs. monetary policy

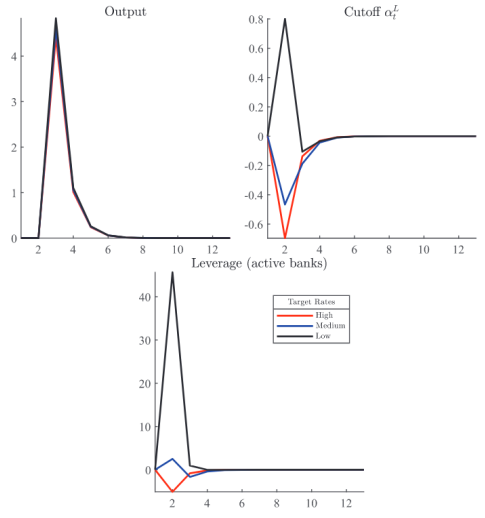


FIGURE 7

Partial equilibrium IRF to a 100 basis points fall in deposit rates. Scale in percentage point deviations from the baseline

$$R_t = R_{t-1}^{\nu} \bar{R}^{1-\nu} \varepsilon_t^R$$

- High r^D: no trade-off
- Low r^D: large intensive margin and exit of the most risk averse intermediary
- Large mispricing of risk: the riskier intermediaries engage the most in risk-shifting

General equilibrium

HHs intertemporal behaviour dictates the supply of loanable funds. Monetary policy as a change in the cost of wholesale funds relative to retail funds.

$k_{i,t}$	$\omega_{i,t}$
$s_{i,t}$	$d_{i,t}$
	$l_{i,t}$

$R_t^L = R_t^D(1-\gamma_t),\ log(\gamma_t) \sim AR(1),\ l_{i,t} = \chi d_{i,t}$

Asset market clearing: $D_t^H = (1+\chi)^{-1} \int_{\alpha_t^L}^{\bar{\alpha}} (k_{i,t} - \omega) dG(\alpha^i)$

Goods market clearing : $S_{t-1}^H + S_{t-1}^I + Y_t = C_t^H + C_t^I + S_t^H + S_t^I + I_t + T_t^L$

Deleveraging phases will typically imply $I_t \leq 0$

The trade-off in GE

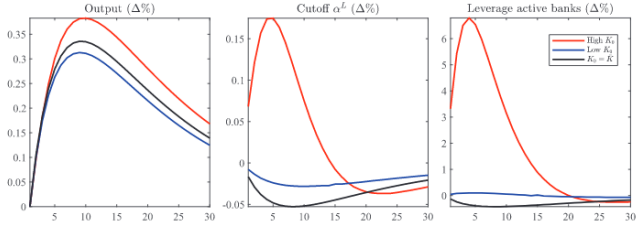


FIGURE 8

Monetary policy shock of 100 basis points to γ_t

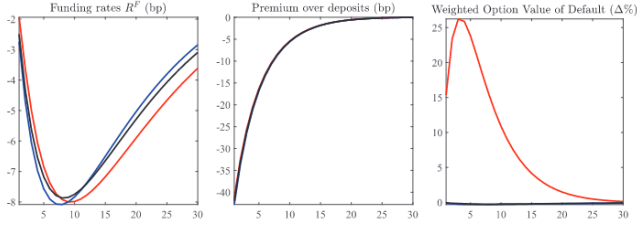


FIGURE 9

Monetary policy shock of 100 basis points to γ_t: financial variables

Credit "supply" boom vs. credit "demand" boom

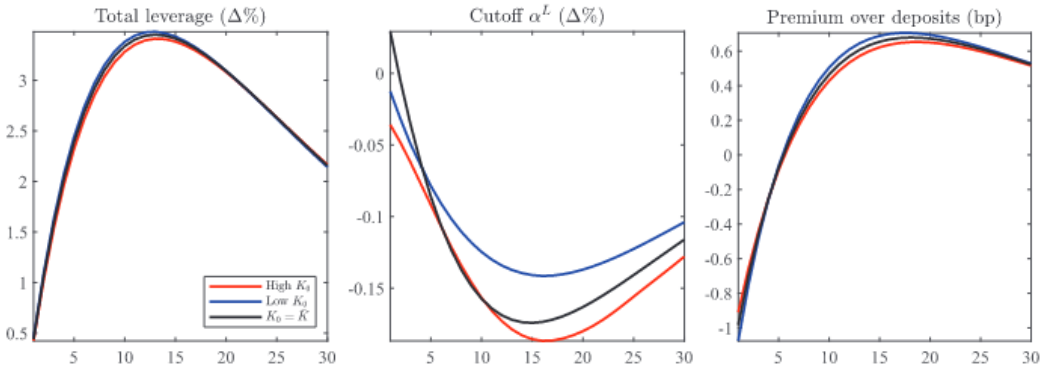


FIGURE 10

Shock to exogenous productivity

Costly intermediary default and contagion

Default now brings a deadweight loss (bankruptcy costs, inaleniable investments) $\bar{\Delta}$. It affects only defaulting intermediaries on impact but may be carried forward to all intermediaries in the future

$$\xi_t \sim \mathcal{B}(1-p)$$
$$\mu_t^d = \frac{\int k_{i,t} \mathbb{1}_{\{\pi_i<0\}} dG(\alpha^i)}{K_t}$$
$$\Delta_t = \xi_{t-1} \max(\mu_{t-1}^d \bar{\Delta}, \Delta_{t-1})$$

ξ_{t-1} is known when intermediaries make their investment decision, so that the main difference is $Z_{t+1}^e = (1-\Delta_t) Z_t^{\rho^Z}$

Large productivity shock

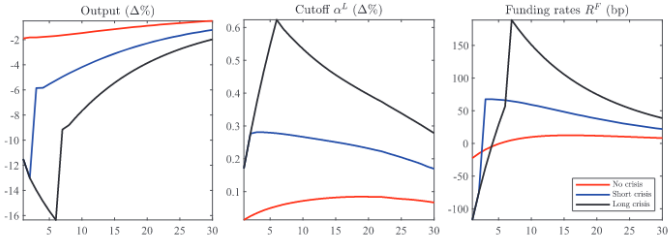


FIGURE 11

Large shock to exogenous productivity

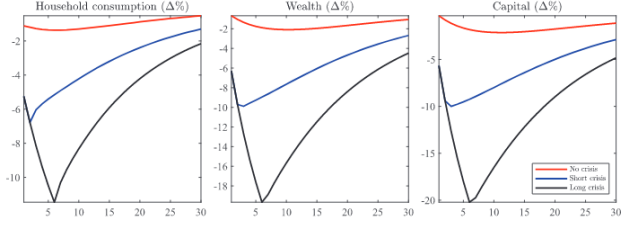


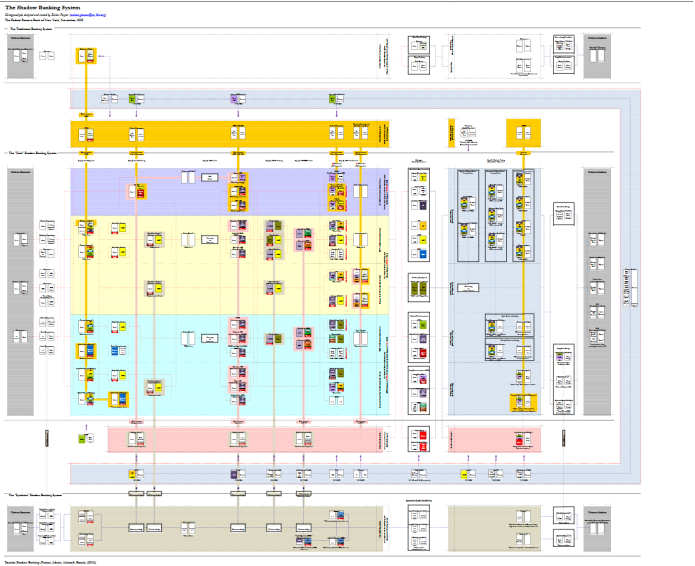
FIGURE A6

Large shock to exogenous productivity: real variables

What I like

- Relatively straightforward and elegant way to implement a financial sector within an otherwise standard macroeconomic model. Many extensions are possible (pricing block through retailers for instance)
- Procyclical leverage
- Relevant w.r.t. "current" institutionnal features
- Can analyse macroprudential and monetary policy separately
- Most macro models with financial frictions take the stock of vulnerabilities as given//here it is the consequence of the build-up phase of the cycle. $\longrightarrow$ relevant change of frame, consequences for overall macrodynamics?

Comment #1 : Finance?



Comment #2 : Risk?

- Relative price of "capital" and associated volatility are not part of the analysis//this is what VaR is mostly about
- Tow-way relationship between total volatility and leverage not analyzed//key-feature of the procyclicality/strategic complementarities associated with VaR (Danielsson et al., 2009, Adrian and Boyarchenko, 2015)
- Very cool to change the frame to look at the build-up phase but “Booms gone bust”? Not quite there yet...

La Fin

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